Physical Security — SY0-701 Study Notes (Section 04)

Maps to exam objectives: 1.2 (compare & contrast types of security controls — the **physical controls** family) and 2.4 (given a scenario, analyze indicators of malicious activity).

Physical security shows up on SY0-701 as **definition discrimination** ("Which control stops a vehicle-borne threat?") and **scenario questions** ("A guard finds a rug thrown over the fence — which brute-force technique is this?"). The single most-tested distinction in this whole section is **piggybacking vs. tailgating** — know it cold.

1. Perimeter Controls: Fencing & Bollards (Obj 1.2)

Fencing — three purposes

1. **Visual deterrent / boundary** — marks where public space ends and private space begins.
2. **Physical barrier** — must be climbed, cut, or dug under; slows an intruder down.
3. **Delay** — buys time for detection and response. Fencing is **adaptable for large perimeters**.

Fence countermeasures — match the attack to the defense (classic exam table)

Attack against fence	Countermeasure
Climbing	Taller fence, **electrification**, **barbed wire** at the top
Cutting	More **robust materials**, **monitoring** (cameras/sensors/guards)
Digging under	**Underground portion** of the fence or **buried mesh**

Bollards

- **Definition:** short, sturdy vertical posts (steel or concrete) that control **vehicle** access.
- **Why they exist:** post-9/11 **anti-terrorism** measure — keep vehicle bombs (IEDs) outside the **blast radius** of a building; also stop **ram raiding** (smashing through storefronts to steal).
- Real-world disguises: **cement planters**, the **red balls outside Target**.
- Standard to know: **ASTM F2656-07 M30 P1** — stops a **15,000-lb vehicle at 30 mph** with penetration of **less than 1 meter (P1)**. The primary threat bollards address is **direct vehicle impact**.
- **Temporary/removable bollards can be tampered with or removed** — a scenario question may ask for the weakness.

Trap — fencing vs. bollards: fencing = **people/perimeter** threats over large areas; bollards = **vehicular** threats in a **specified area**. "Vehicle-borne threat near the building entrance" → bollards, every time.

2. Brute Force Attacks (Obj 1.2 / 2.4)

Two meanings you must separate:

- **Cyber/crypto:** trying every possible combination until one works. Example from the course: a **4-digit PIN has 10,000 combinations** (0000–9999).
- **Physical:** forcible entry. Four forms:

Physical brute-force form	What it looks like	Defense
Forcible entry	Breaking down doors; **glass cutting** to defeat windows	High-strength doors, **deadbolts**, **metal frames**, **solid-core** doors; **reinforced/laminated** glass
Tampering with security devices	Forcing a gate open; **blinding cameras/sensors** with paint, lights, or mirrors; smashing sensors	**Redundancy / multiple layers** of controls
Confronting security personnel	Distraction, assault, weapons	Guard training in **conflict resolution & self-defense**, **quick backup communication**
Ramming barriers with vehicles	Driving into government/military/bank buildings	**Bollards** and reinforced barriers that **absorb/dissipate energy**

The course's color details that make great distractors: a **rug thrown over barbed wire** to climb a fence (still brute force — climbing/cutting/ramming fences), and **the lock is the weakest point of a door**.

3. Surveillance (Obj 1.2)

Definition: an organized strategy/setup to **observe and report** activities. Four categories:

a) Video surveillance (CCTV)

- **Wired:** cabled to a monitoring station — harder to interfere with.
- **Wireless:** easier to install, but susceptible to **jamming/interference**.
- A **detective control** (records what happened; does not prevent).
- Features: **motion detection, night vision, facial recognition, remote access**; **PTZ (Pan-Tilt-Zoom)** cameras operated from a **central guard station**.
- Placement: **entrances and exits** of critical infrastructure — **data centers, communications closets, building doors**.
- Process: **record + review** footage.

b) Security guards

- The **most flexible and adaptable** surveillance/control tool.
- Can **detect unusual behavior, respond immediately, make judgment calls**, and act as a **visual deterrent**.
- Roles at checkpoints: **visual deterrent, verification assistance** (e.g., when the badge system fails/offline, visitors without badges → identity check + **temporary badge**), and **immediate response** (confront intruder or initiate emergency protocols).

c) Lighting

- Crucial because it makes **video and guards more effective**: deters criminals, **reduces shadows/hiding spots**, improves footage quality.

- **Motion-activated lights** = deterrent **and** alert in one.

d) Sensors — know the four types cold

Sensor	How it works	Exam hook
Infrared	Detects **warm bodies** / heat	Excellent in **low light or darkness**; can be **defeated by heating the room**
Pressure	Detects **weight** on a floor area or mat	Simple contact/weight trigger
Microwave	Emits microwave pulses, measures reflection	Covers **large areas**; **prone to false alarms at high sensitivity**
Ultrasonic	**Echolocation** (like a bat)	Used in **automated doors**, indoor use

4. Bypassing Security Controls (Obj 2.4)

This is the 2.4 content — "analyze indicators of malicious activity." The exam describes an odd observation and asks what attack it indicates.

Bypass method	Techniques	Countermeasure
Visual obstruction	**Spray paint, foam, stickers/tape** over a camera lens; **balloons/umbrellas** held over it	Tamper alarms, guard response
Blinding sensors/cameras	**Flashlights or lasers** aimed at the lens; **heating the room** to blind **infrared** sensors	Redundant sensor types, tamper alarms
Interfering with acoustics	**Loud music, jamming devices, white-noise machines**	—
Interfering with electromagnetics	**EMI / jamming** wireless setups; EMI generators; targeting the **Wi-Fi band** in a no-wireless facility	**Encrypted frequency hopping** against jamming/eavesdropping
Attacking the physical environment	**Temperature changes** to trick sensors; **killing the power supply** (unplugging, tampering with power boxes, sabotaging transformers); **arson** as distraction/destruction	**UPS / backup power**, tamper alarms
Physical tampering	**Cutting wires** (crude but effective; requires close proximity)	Tamper alarms, layered controls

5. Access Control Vestibules & Entry Attacks (Obj 1.2)

The vestibule (a.k.a. mantrap)

- A **double-door system**, electronically controlled so that **only one door can be open at a time**.
- Creates an **intermediate space between the untrusted outside area and the trusted inner area** — a person who fails verification is **trapped between the doors** until security personnel intervene.
- Often combined with **badges + security guards** for a multi-layered framework.
- Single-person-sized vestibules **prevent piggybacking and tailgating** (no room for two).

Piggybacking vs. Tailgating — THE trap of this section

	Piggybacking	Tailgating
Who is complicit?	The **authorized person intentionally allows** the unauthorized one in — through **complicity or negligence**	**No consent** — the authorized person doesn't know
How?	Usually **social engineering** (course example: pentester dressed as a **delivery driver with boxes** asking an employee to badge him in — worked ~50% of the time)	**Opportunistic** — closely following someone through a door/turnstile without their knowledge
Nature	Collaborative/deceptive	Covert

Access badges

- Embedded tech: **RFID** (radio frequency identification), **NFC** (near-field communication), legacy **magnetic stripe**.
- Cross-referenced against a **centralized database**; grants/denies based on the person's **role-based access level** (course example: NSA — building access ≠ individual office access).
- Every badge use is **logged → audit trail** for breach review.

6. Door Locks (Obj 1.2)

Core idea: all perimeter defenses become useless the moment someone is **inside the building** — interior locks protect the sensitive spaces (server rooms, network closets, offices).

Weak vs. strong locks (course timing = great distractors)

- **Padlock** (basic **pin-and-tumbler**): defeatable in **~15 seconds** with a **lock pick + tension wrench**.
- **Privacy door locks** (bedroom/bathroom): pop open with a **slim rod/needle**, or a **flathead screwdriver/coin** on the slotted type.
- **Traditional keyed door locks:** defeated in **~30–60 seconds** by skilled attackers.

Electronic lock mechanisms

Mechanism	Notes
PIN / identification number	Course example: **8-digit PIN = 1 in 100 million** guess chance; per-person PINs enable **logging/auditing**
Wireless signal	**NFC tap** (smartphone), plus **Wi-Fi, Bluetooth, RFID** locks
Biometric	Fingerprint, **retinal scan**, **facial recognition** = **inherence factor — "something you are"**
Cipher lock	Older **mechanical** push-button **combination** lock; used on **server rooms/network closets**; costs more, higher protection

Biometric accuracy — FAR / FRR / CER (heavily tested)

- **FAR (False Acceptance Rate):** system accepts someone who **should NOT** be granted access. Drive toward zero by **increasing sensitivity**.

- **FRR (False Rejection Rate):** system **denies someone who SHOULD** have access — just as big a problem (authorized people locked out, half the time in the course example).
- **CER / EER (Crossover/Equal Error Rate):** the point where **FAR = FRR**. **The lower the CER, the better the biometric system.** When buying a fingerprint lock, compare CERs.

MFA at the door

- Badge (something you **have**) + PIN (something you **know**); or **PIN + fingerprint**.
- Example from the course: open lobby → **turnstiles** requiring **badge + PIN** → the lobby/turnstile area itself is an **access control vestibule**.

7. Access Badge Cloning (Obj 2.4)

The attack — four steps

1. **Scanning/reading:** handheld RFID/NFC reader captures the badge data — can be **concealed in a backpack**; done **without the owner's knowledge**.
   - Ranges: **NFC ~1–2 inches; RFID ~2–10 inches** (stronger antenna can roughly double this — attacker still must get close).
2. **Data extraction:** pull the **unique identifier / encrypted data** from the capture (can happen later, anywhere).
3. **Writing:** transfer to a **blank card/device**. Course tool: the **Flipper Zero** — reads **125 kHz RFID**, stores many codes, then **emulates** the badge or writes a physical clone.
4. **Using:** the clone behaves exactly like the original — building access, systems, even **NFC payment fraud**.

Why it's dangerous: **easy, cheap, stealthy** (no need to possess the original badge), and compromises physical security at scale.

Defenses (six — matching/multi questions love this list)

1. **Advanced encryption** on the badge data (many basic systems use simple identifiers with **no robust encryption**).
2. **MFA** — badge + PIN/password/biometric; a cloned badge alone is then useless.
3. **Regularly update protocols/encryption keys** so clones stop working.
4. **User education** — cautious storage, report suspicious access attempts.
5. **Shielded wallets/sleeves** (RFID shielding) so the badge can't be read while carried.
6. **Monitor & audit access logs** — detect **impossible usage** (course example: badge used at the office while the owner is at a conference across the country).

Rapid-Fire Self-Test

1. Fence attack: climbing → ? Cutting → ? Digging → ? (taller/electrified/barbed • robust materials+monitoring • buried portion/mesh)
2. Bollard standard ASTM F2656-07 **M30 P1** means? (15,000 lb @ 30 mph, <1 m penetration)
3. Fencing vs. bollards — which threat does each address? (people/perimeter vs. vehicles)
4. A 4-digit PIN = how many combinations? (10,000)

5. Weakest point of a door? (the lock)
6. Defense vs. ramming vehicles? (bollards/reinforced barriers that absorb/dissipate energy)
7. Which sensor uses echolocation, and where do you see it? (ultrasonic · automated doors)
8. Which sensor covers large areas but false-alarms at high sensitivity? (microwave)
9. How do you blind an infrared sensor? (heat up the room)
10. Countermeasure vs. wireless jamming/eavesdropping? (encrypted frequency hopping)
11. Define a vestibule. (double doors, only one open at a time)
12. Piggybacking vs. tailgating in one line each. (authorized person intentionally lets someone in / unauthorized follows without consent)
13. FAR vs. FRR? Which way do you tune sensitivity? (accepts bad / rejects good · ↑ sensitivity lowers FAR, raises FRR)
14. What is the CER and do you want it high or low? (FAR=FRR crossover · lower is better)
15. NFC vs. RFID cloning ranges? (1–2 in / 2–10 in)
16. Name the Flipper Zero's two ways to use a captured badge. (emulate / write to blank card)
17. Clone used while the real owner is across the country — which control catches it? (access-log monitoring/auditing)
18. A guard station with PTZ cameras — what does PTZ stand for? (Pan-Tilt-Zoom)
19. Which lock is mechanical, push-button, combination-based? (cipher lock)
20. Motion-activated lights give you what two benefits? (deterrent + alert)

Passing reminder: the real SY0-701 needs **750/900** — expect 1.2 physical-control questions and 2.4 "indicator of malicious activity" questions to come largely from Lessons 30, 32, and 35.